Chapter 3 Part II - Cost Assignment / Allocation – Normal Costing

Learning Objectives

Why do we need to allocate manufacturing overheads?

How to apply manufacturing overheads?

Compute a predetermined manufacturing overhead rate and use it to allocate MOH to jobs

Determine the cost of a job and use it to make business decisions

Compute and dispose of over allocated or under allocated manufacturing overhead

Why do we need an allocation base for manufacturing overhead?

It is impossible or difficult to trace overhead costs to particular jobs.

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Manufacturing overhead consists of many different items ranging from the grease used in machines to a production manager’s salary.

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Actual overhead for the period may not be known until the end of the period.

How to apply Manufacturing Overheads?

Step 1: Compute the Predetermined Overhead Rate

Before the accounting period starts. Based on ESTIMATED data.

Step 2: Apply Manufacturing Overhead to Products or Jobs

During the accounting period, based on ESTIMATED and ACTUAL data.

Step 3: Reconcile the Difference between Actual and Applied Overhead.

At the end of the accounting period, based on ACTUAL and APPLIED.

Note that Estimated and Applied DO NOT mean the same thing. Applied is a combination of estimated and actual.

How to apply manufacturing overheads – cost assignment – Normal Costing

Step 1: Compute the Predetermined Overhead Rate

$$\text{Predetermined Overhead Rate} = \frac{\text{Estimated Total Manufacturing Overhead Cost}}{\text{Estimated Total Units of the Allocation Base (Cost Driver)}}$$

Some Possible Cost Drivers:

1. **direct labor-hours**
2. **machine-hours**
3. **direct labor dollars**

Step 2: Apply Manufacturing Overhead to Products (Jobs) as they are produced

Based on estimates, and determined before the period begins.

Predetermined Overhead Rate × Actual Value of the Cost Driver (for a specific job) = Overhead Applied to an Individual Job

Actual amount of the cost driver such as direct labor hours, machine hours, # of set-ups, etc.

Step 3: Reconcile the Difference Between Actual and Applied

Actual Overhead > Applied Overhead = Under Applied

Actual Overhead < Applied Overhead = Over Applied

Why are actual overheads different from applied overheads?

What do we do with any over/under applied overheads?

Adjust Under and Over Applied Overheads

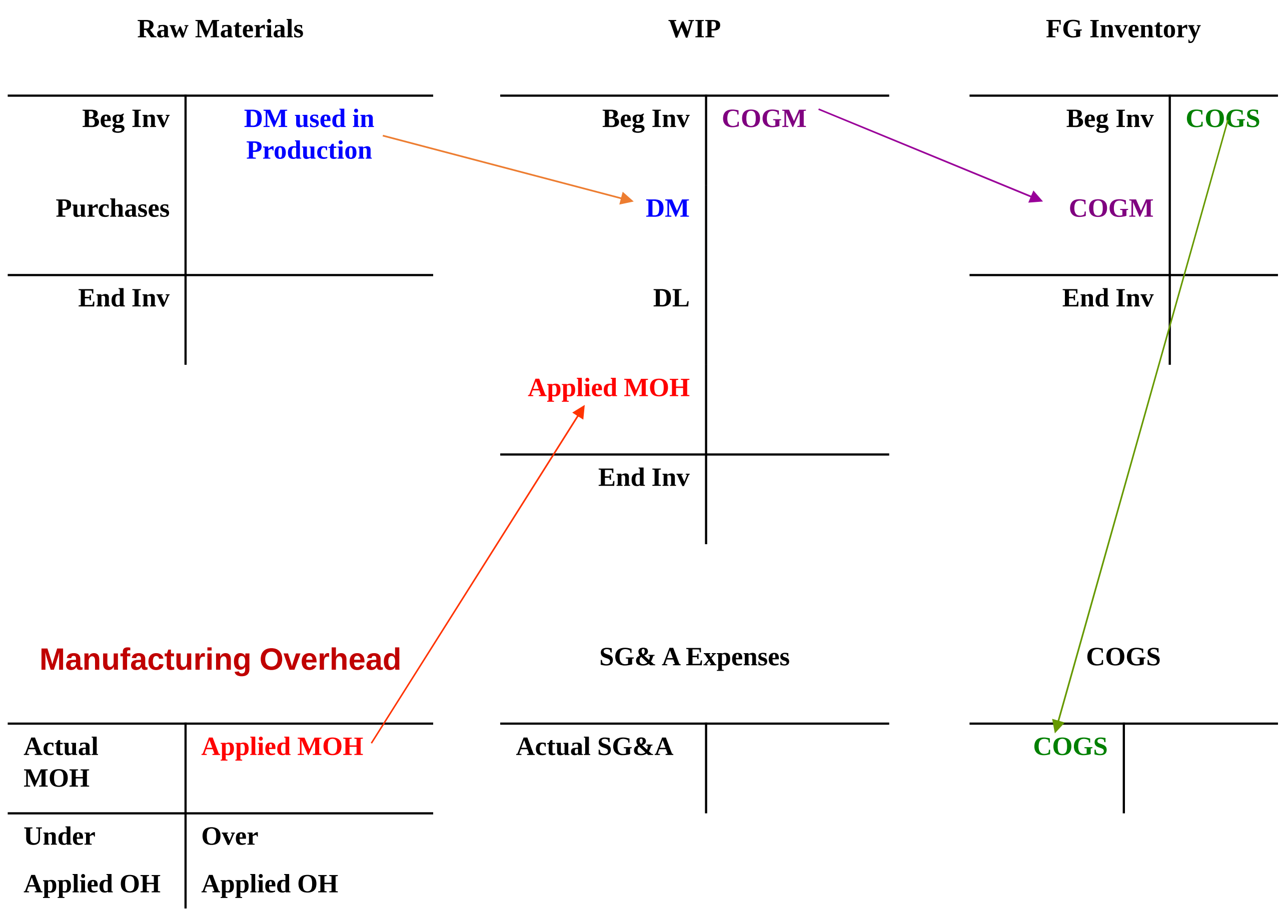
The *manufacturing overhead applied* to jobs during a period is almost never equal to actual factory overhead costs incurred during that period:

If Manufacturing Overhead is . . .	Adjust to Cost of Goods Sold
UNDERAPPLIED (Applied OH is less than actual OH)	INCREASE Cost of Goods Sold
OVERAPPLIED (Applied OH is greater than actual OH)	DECREASE Cost of Goods Sold

Underallocated or Overallocated Manufacturing Overhead

- Underallocated (undercosted)
  - – not enough allocated to jobs
  - – too little expense
- Overallocated (overcosting)
  - – too much allocated to jobs
  - – too much expense

T-ACCOUNTS OF INVENTORY FLOWS USING NORMAL COSTING



COGM/COGS using Normal Costing

Raw Materials	Beginning RM + Purchases - Ending RM
Work in Process	DM used in Production + DL + Overhead - **Applied** Total Manufacturing Costs + Beginning WIP - Ending WIP
Finished Goods	**Cost of Goods Manufactured** + Beginning FG - Ending FG
	= Cost of Goods Sold

This is exactly the same COGM/COGS schedule we did in Chapter 2-
the ONLY difference is that MOH is now MOH APPLIED, NOT actual MOH!

Application of Manufacturing Overhead Numerical Example

Delta Company applied manufacturing overhead to products on the basis of Direct Labor Hours

Estimated (budgeted) manufacturing overhead costs were $900,000, and total estimated direct labor hours are 50,000.

Actual manufacturing overhead costs were $925,000 and actual direct labor hours were 55,000.

1. What is the predetermined overhead rate?
2. How much overhead was applied?
3. How much was over/underapplied?

Application of Manufacturing Overhead Numerical Example

Delta Company applied manufacturing overhead to products on the basis of DL Hours

Estimated (budgeted) manufacturing overhead costs were $900,000, and total estimated direct labor hours are 50,000.

Actual manufacturing overhead costs were $925,000 and actual direct labor hours were 55,000.

1. What is the predetermined overhead rate? 900,000/50,000 = 18/DL hour

2. How much overhead was applied? 18*55,000 = 990,000

3. How much was over/under applied? 990,000-925,000 = 65,000 over applied